

Smart wine cellar

Diogo Pisco ist1115027

Beatriz Coelho ist1107113

Instituto Superior Técnico
Group 25

Description of the project

Smart Wine Cellar is an IoT system designed to monitor and supervise wine storage conditions in real time. The prototype measures temperature, humidity, air quality, and wine level using sensors connected to an Arduino MKR NB 1500. The collected data is transmitted through the mobile network using MQTT, forwarded to Firebase through a Python bridge, and displayed in an Android application. The system also supports ventilation control and warning generation, helping maintain adequate environmental conditions for wine preservation.

Commercial products

- Govee WiFi Thermo-Hygrometer

Remote monitoring of temperature and humidity.



- SensorPush Wireless Sensor

Real-time monitoring of temperature and humidity, with app alerts and data history.

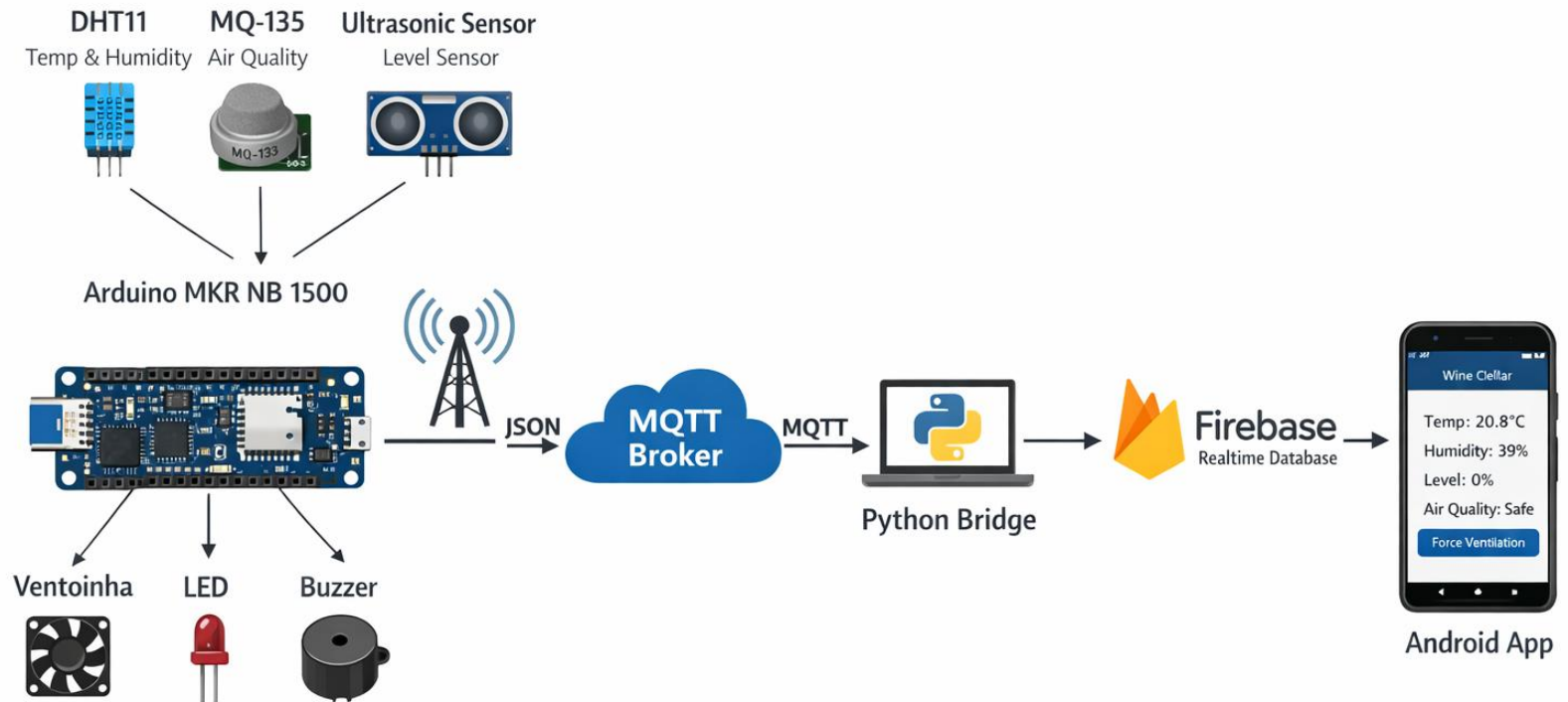


- Wine Guardian

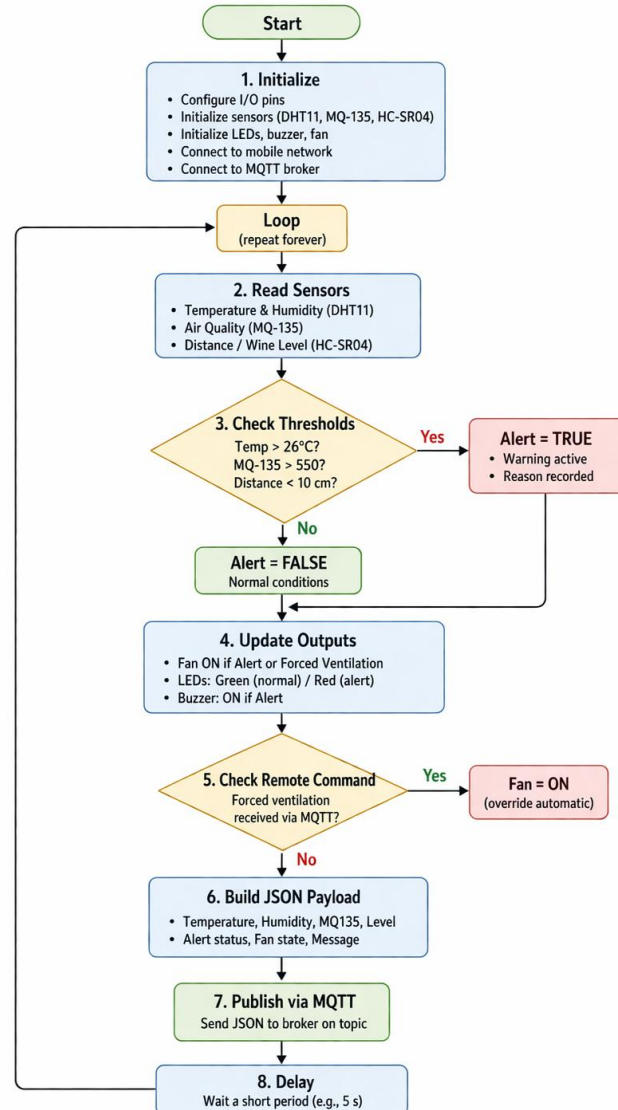
Wine cellar climate control solution.



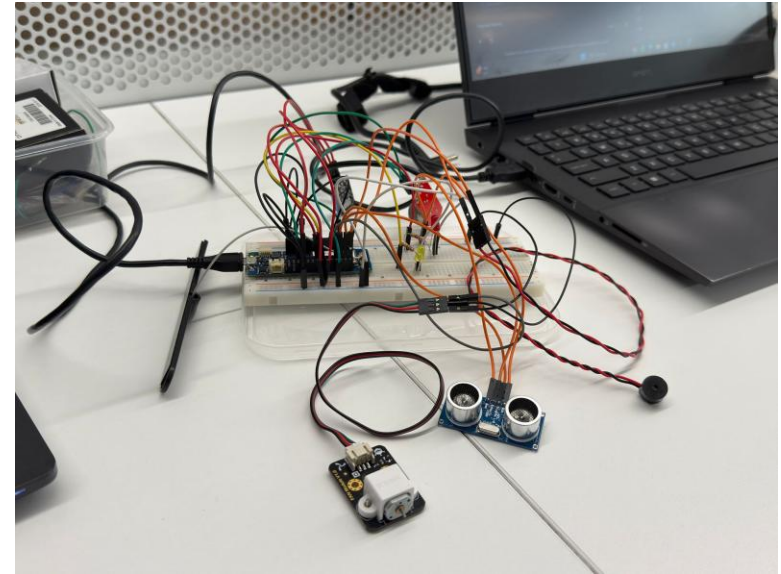
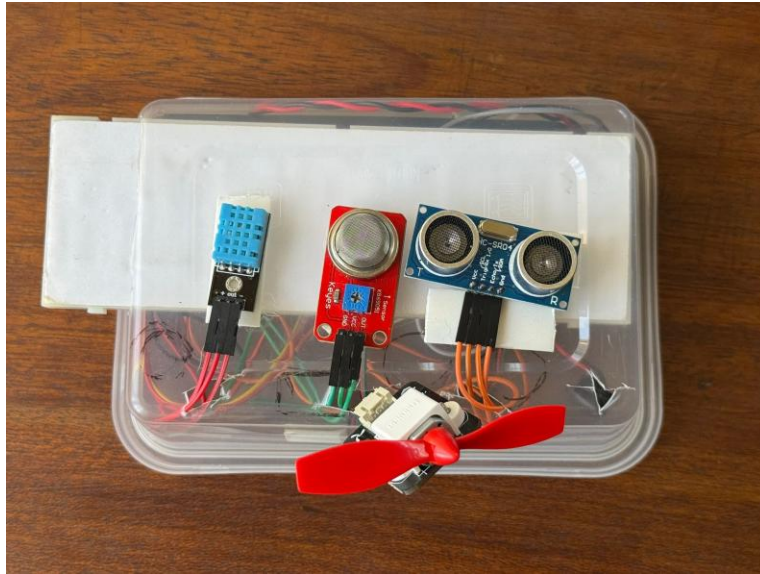
Overall architecture



Algorithm executed at the MCU



Hardware assembly



Database platform



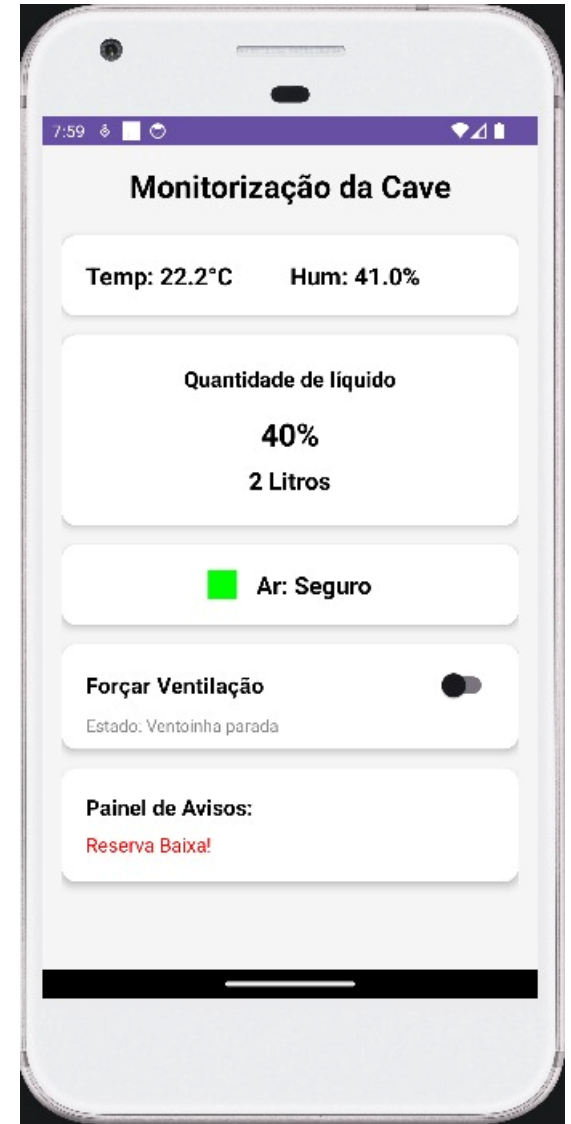
Firebase

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https://monitorizacao-cave-default-rtdb.europe-west1.firebaseio.com/
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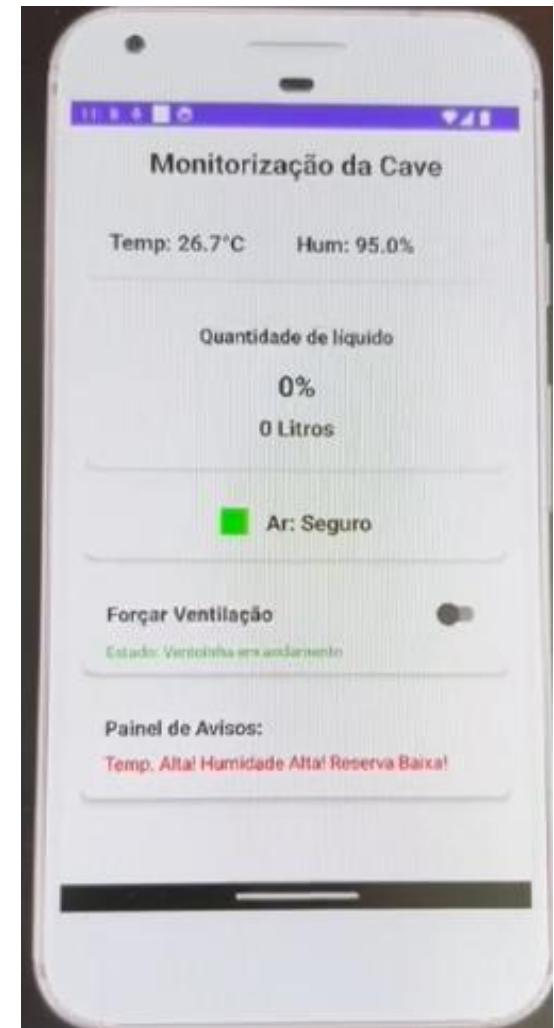
```
— Avisos: "Reserva Baixa! "  
— Humidade: 41  
— Nivel_Cuba_Litros: 2  
— Nivel_Cuba_Percentagem: 40  
— Qualidade_Ar_Segura: true  
— Temperatura: 22.2  
— Ventoinha_Estado: "Ventoinha parada"
```

App functionalities

- Real-time monitoring of temperature, humidity, air quality, and wine level
- Fan control and fan state display
- Warning messages for abnormal conditions



Activities of the Mobile App



Conclusion

The developed system enables real-time monitoring of temperature, humidity, air quality, and wine level, while supporting automatic and remote ventilation control through an integrated IoT architecture.

Possible improvements include better wine-level calibration, more accurate sensors, push notifications for alerts, and direct cloud integration without a PC-based bridge.